

PlantDoc Dataset

PlantDoc is a dataset provided by indian researchers through the "*PlantDoc: A Dataset for Visual Plant Disease Detection*" paper.

It is composed by **2598** images across **13** plant species and **27** classes.



Real-world context and problem relevance

Food Needs

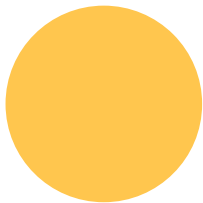
According to the *Food and Agriculture Organization of the United Nations*, the world will require **50%** more food by **2050** due to the increasing population.

Crops Lost

Every year, up to **40%** of the crops are lost due to **plant pests** and **diseases**.

Reference: FAO

Domain Challenges

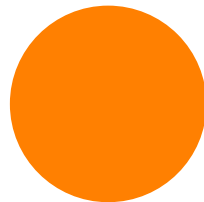


Hardware Cost & Lack of expertise

Sensors have been the most frequent tool used to monitor plants health but it is an **expensive** solution that also requires a certain degree of **expertise**.

Nature of the Data

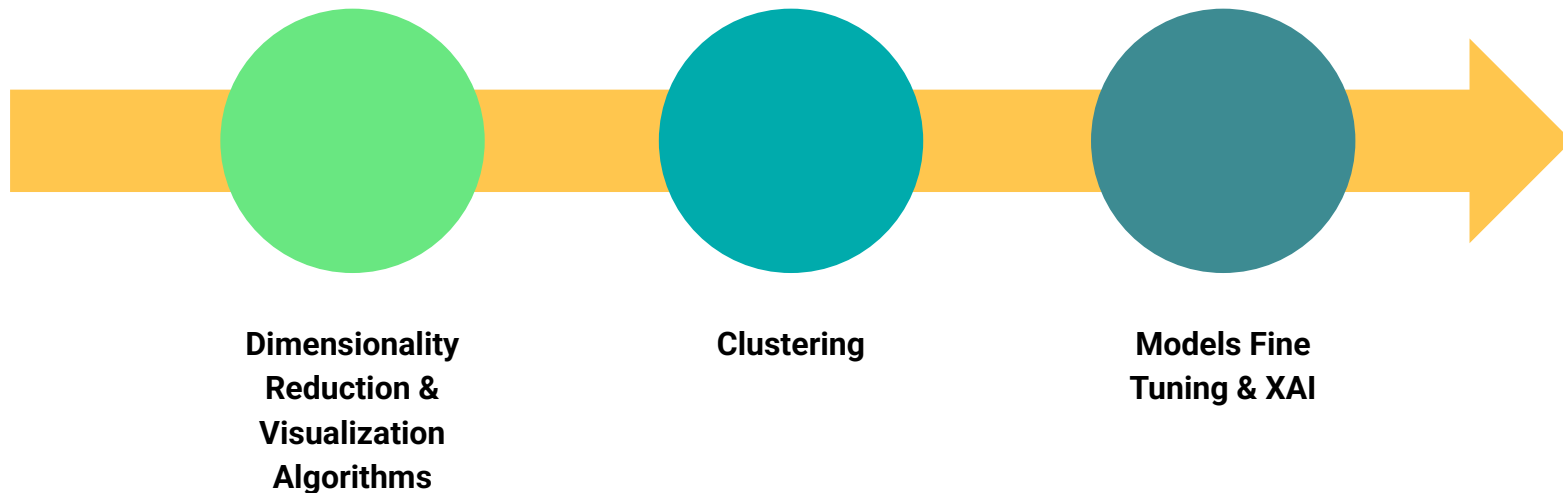
Most of the images obtained in contexts relative to this problem are **not lab-controlled**: images don't have plants with perfect shape, same object size, perfect disease manifestation, same backgrounds, etc.



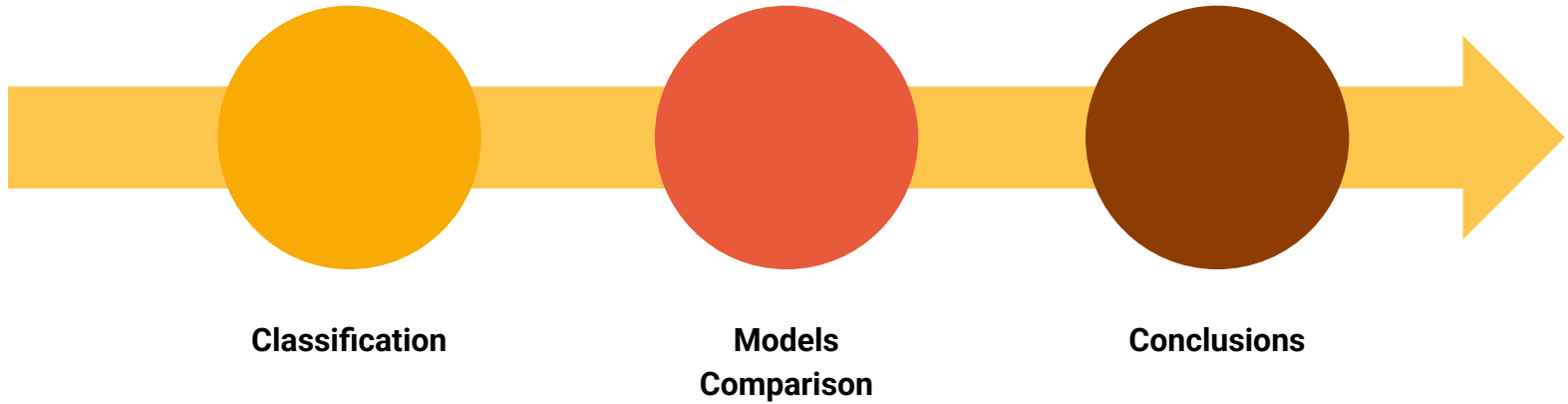
Analysis Pipeline



Analysis Pipeline



Analysis Pipeline



PlantDoc Dataset samples

	Apple	Bell Pepper	Blueberry
Normal			
Rust			
Scab			

Corn	Grape	Peach
		
		

Squash	Strawberry	Tomato
		

Feature Engineering

The dataset is entirely composed by **images** and as such they are composed by **pixels** which are not ideal to work with directly for goals like **understanding** and **exploring** data.



Mode	Categorical nominal
Width	Numerical discrete
Height	Numerical discrete
Aspect Ratio	Numerical continuous
Mean Intensity	Numerical continuous
Std Intensity	Numerical continuous
Name	Categorical nominal
Disease	Categorical nominal
Full	Categorical nominal

Data Exploration & Understanding

1

Classes Distribution

2

Channels

3

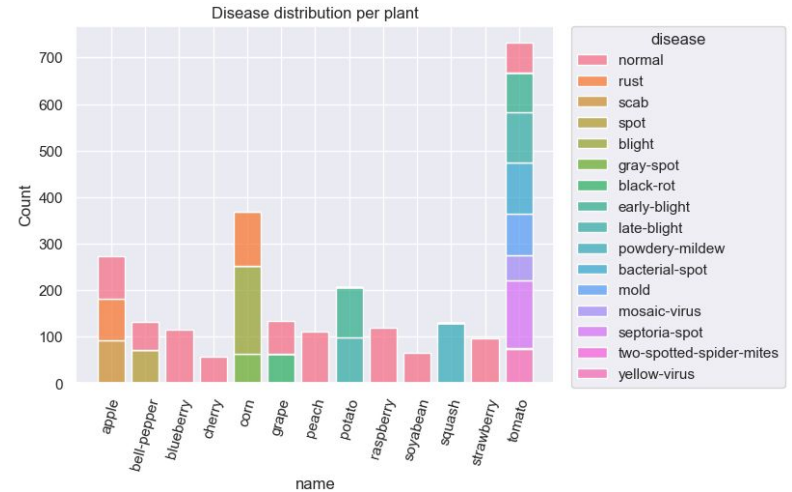
Aspect Ratio

4

Pixel Intensity

Classes Distribution

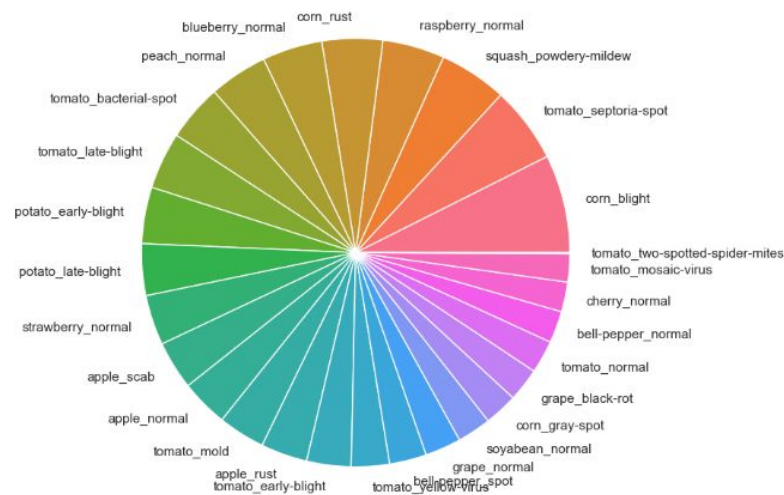
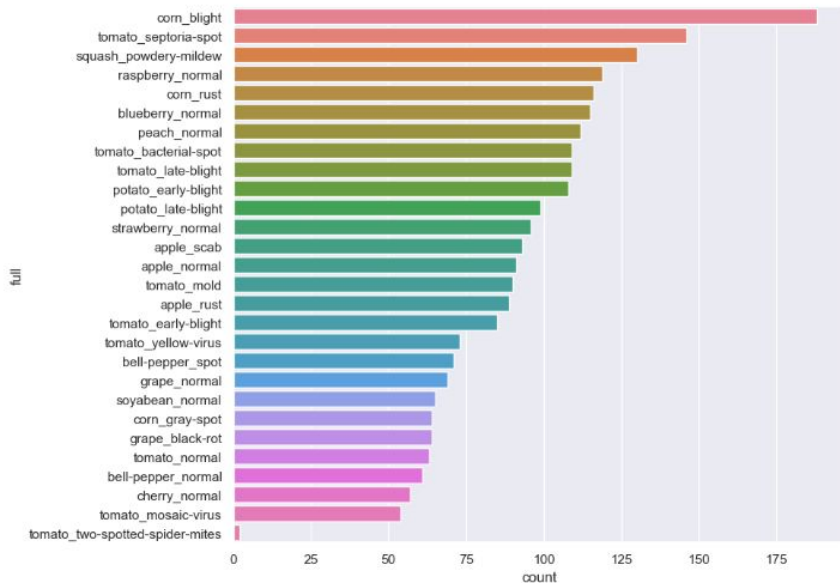
- Some plants like **Corn**, **Potato** and **Squash** are only available through disease examples
- Not all plants are affected by all diseases



Classes Distribution

Class imbalance is present with roughly a **4:1** ratio between the most and least frequent classes

Images distribution by plant+disease



Classes Distribution

Structural Zero

Not all plants are equally affected by same **pests** and **diseases**.

Classes Imbalance

Classes imbalance is going to be dealt with during the **classification** process in two ways

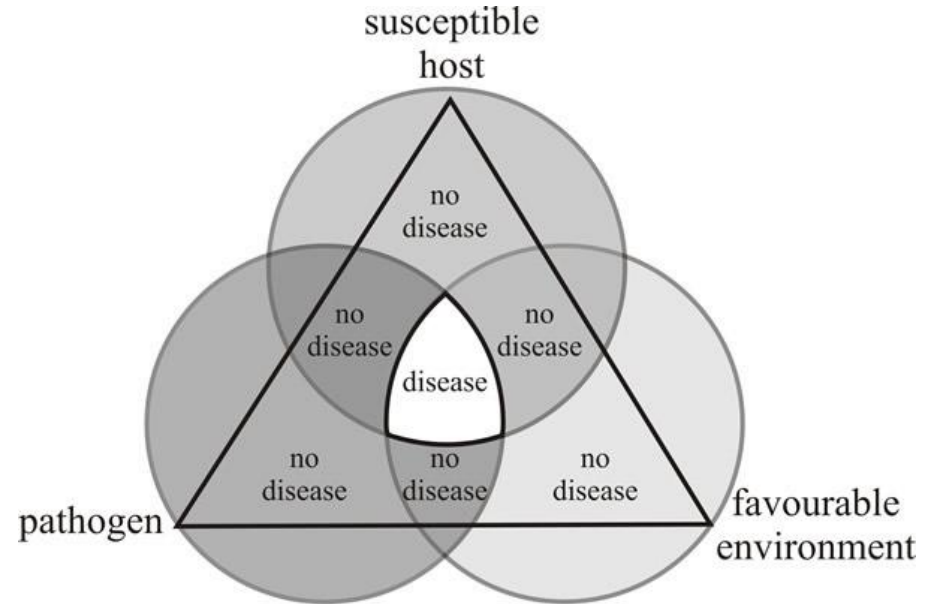
- **Data Augmentation** during the **fine tune** process of the models
- Employing **F1-score** metric

Structural Zero

In **Plants biology**, the concept of **Disease Triangle** partially explains the complete absence of certain classes.

According to the Disease Triangle, there must be three conditions **simultaneously** in order for a plant to get infected:

- *Susceptible Host*, the plant must be of a type that such that the pathogen is **biologically capable** of infecting
- *Suitable Environment*, the weather conditions such as **temperature, humidity** and **moisture** must be right for that pathogen to grow
- *Correct Pathogen*, the specific pathogen that causes that specific disease **must be present**



Structural Zero

The Disease Triangle doesn't explain the absence of **healthy-state** plants like Corn, Potato and Squash.

Two possible approaches to deal with the classification process:

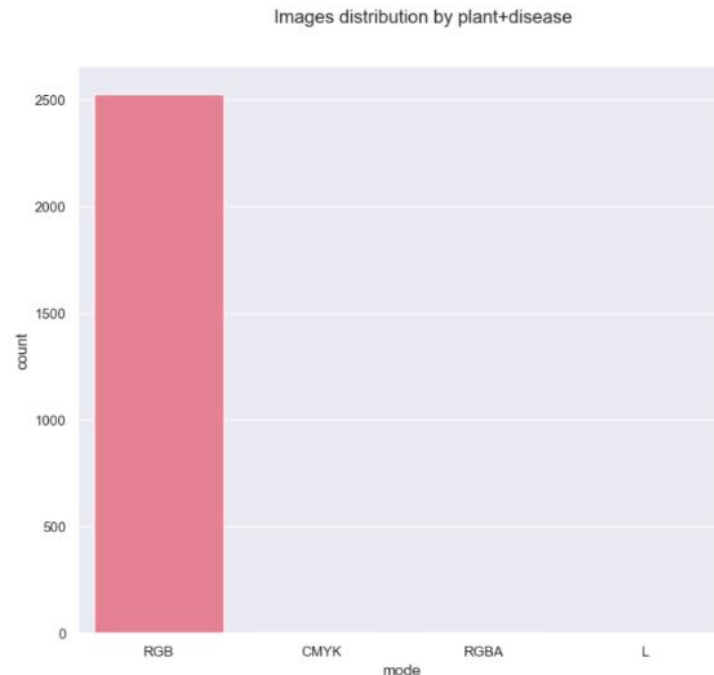
- **Hierarchical**, a first-layer model identifies which of the **13** plant species it is and a second-layer model, for each plant, identifies which disease the plant is affected by.
 - Pros:
 - Moves the classes imbalance problem **locally** to each plant, potentially removing it for those plants that do not suffer from it
 - Cons:
 - **Poor Scalability**, 1 first-layer model and 1 second-layer model for each plant, **14** total
 - **Error Propagation**, first-layer misclassification will generate a guaranteed mistake in second-layer
- **Flat**, a unique model that identifies which of the **27** classes *plant+disease* it is.
 - Pros:
 - **Great Scalability**, 1 model total is required
 - **No Error Propagation**
 - Cons:
 - Moves the classes imbalance **globally**

Channels

Almost **all** images are in **RGB** mode.

CMYK and **RGBA** are converted to **RGB** while **L(greyscale)** are dropped.

Color is a fundamental feature for the problem at hand and therefore images not providing this information lose most of their value



Aspect Ratio

Most of the images present an aspect ratio that is close to 1.

Outlier images are being cropped in order to bring their aspect ratio closer to 1.

Proportion is another crucial property for the problem at hand since a disease could be mistaken for another if its shape were overly distorted.

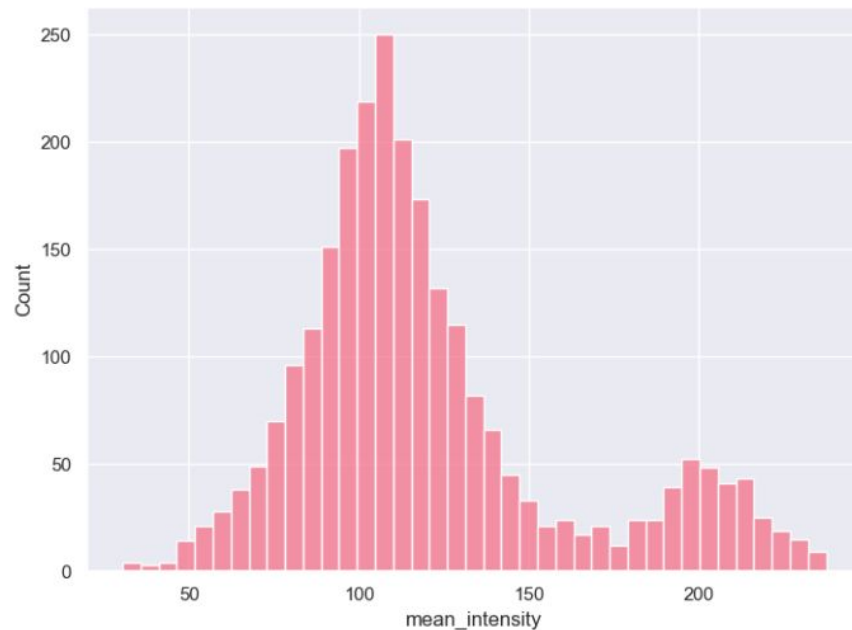


Pixel Intensity

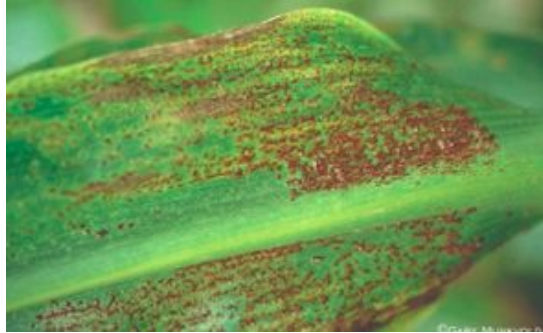
The *mean_intensity* histogram appears to have **two** peaks, being **~110** and **~200**.

There could be two possible causes for this:

- Some images are taken under **different lighting condition**, more light at ~200 and less at ~110
- Some images are taken in **real-world** context (~110) where the background of the subject plant would be the soil and/or other plants keeping the mean intensity of the image low due to the **brown** and **green tones** while other images are **lab-controlled** (~200) where the subject plant has just a **white** background.



Pixel Intensity, ~110



Pixel Intensity, ~200



Embeddings Extraction

1

ResNet-50

2

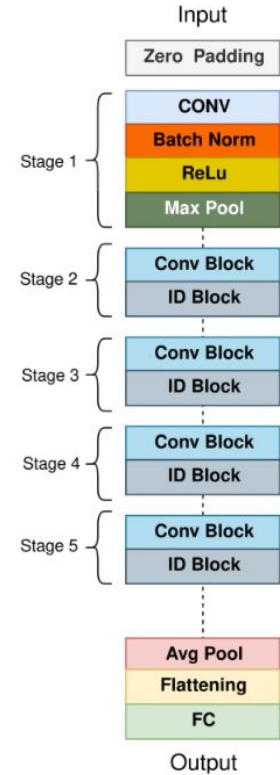
**Google ViT Base
Patch-16-224**

ResNet-50

ResNet-50 is a **Convolutional Neural Network**, *CNN*, that introduces improvements such as **residual learning** and **batch normalization**.

This model has been trained on **ImageNet-1k**.

In this CNN, embeddings are extracted from the last **fully connected layer**, FC, after the **global average pooling** has taken effect and the **spatial properties have been flattened** into a **2048-values** vector.

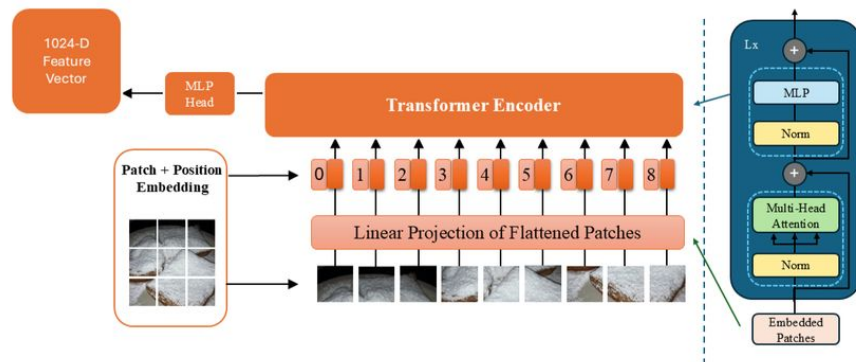


Google ViT Base Patch-16-224

Google ViT Base Patch-16-224 is a **Visual Transformer**, ViT, a model that approaches images similarly to text that is by treating it as a sequence of **patches**.

This model has been trained on **ImageNet-21k**.

In this ViT, embeddings are extracted from the **CLS token** that for the problem at hand serves a somewhat similar purpose of the FC layer, providing a **768**-values vector where informations on the image are compressed.



Dimensionality Reduction & Visualization Algorithms

1

PCA

2

t-SNE

3

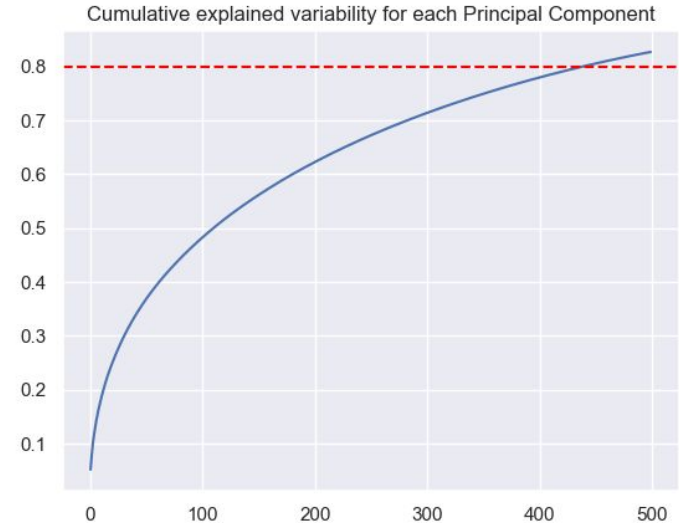
UMAP

PCA

From both ResNet-50 and Google ViT have been extracted a quite high dimensional embedding of the input, respectively **2048** and **768** values.

PCA has been employed to both reduce redundancy and provide **dimensionality reduction**, to respectively **500** and **200** dimensions, while providing at least **80%** variability.

Data has been **standardized** before applying PCA.



t-SNE and UMAP

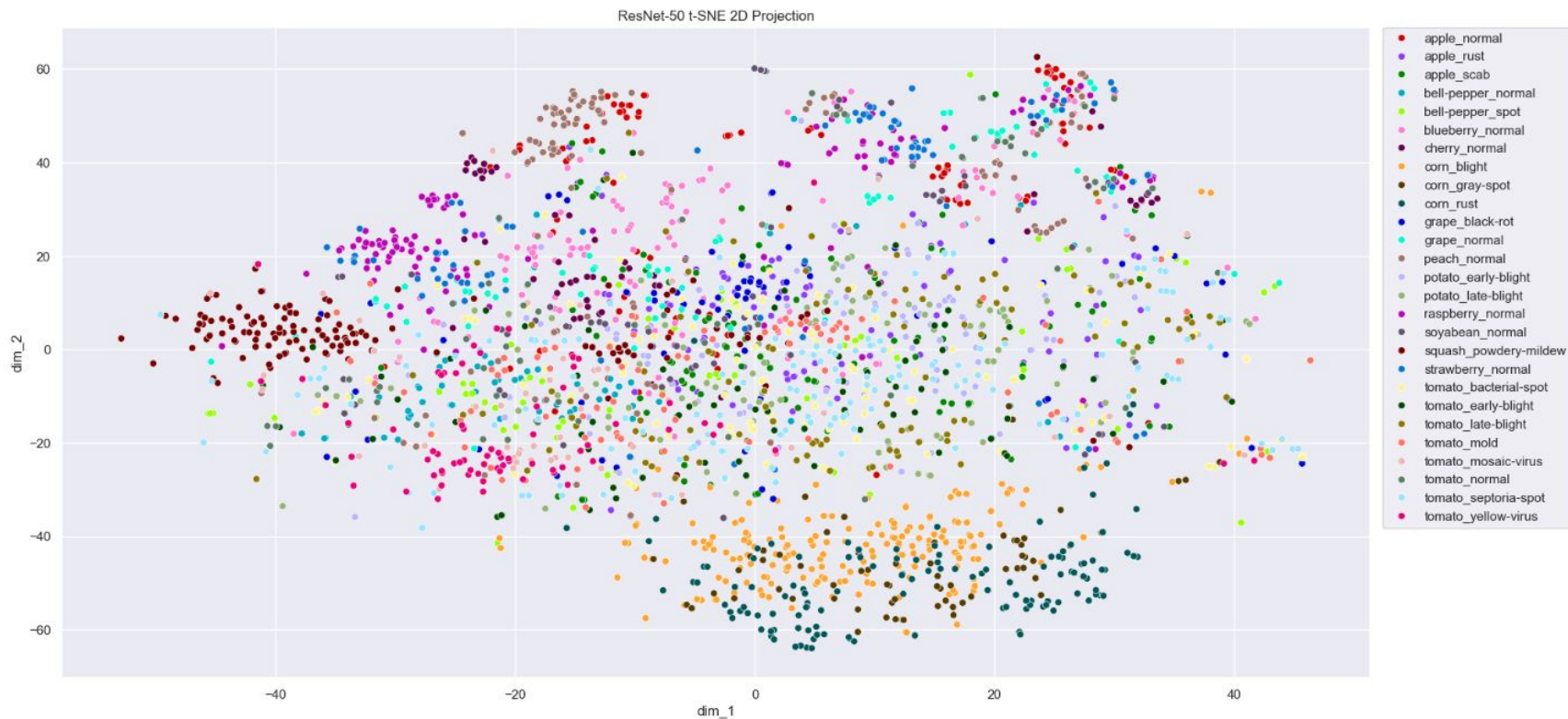
t-SNE and UMAP are **manifold learning** algorithms that allows the representation of high-dimensional complex data in lower dimensions, usually **2** or **3**.

Both algorithms are characterized by several hyperparameters that need to be tuned in order to obtain the best possible representation.

The optimization step has been dealt with through **optuna**, a tool that provides hyperparameters optimization through a Bayesian Optimization approach.

The metric employed together with optuna in order to find the best setup for the two algorithms is **trustworthiness**, a measure of how accurately the low-dimensional representation of high-dimensional data preserves the local neighborhood structure.

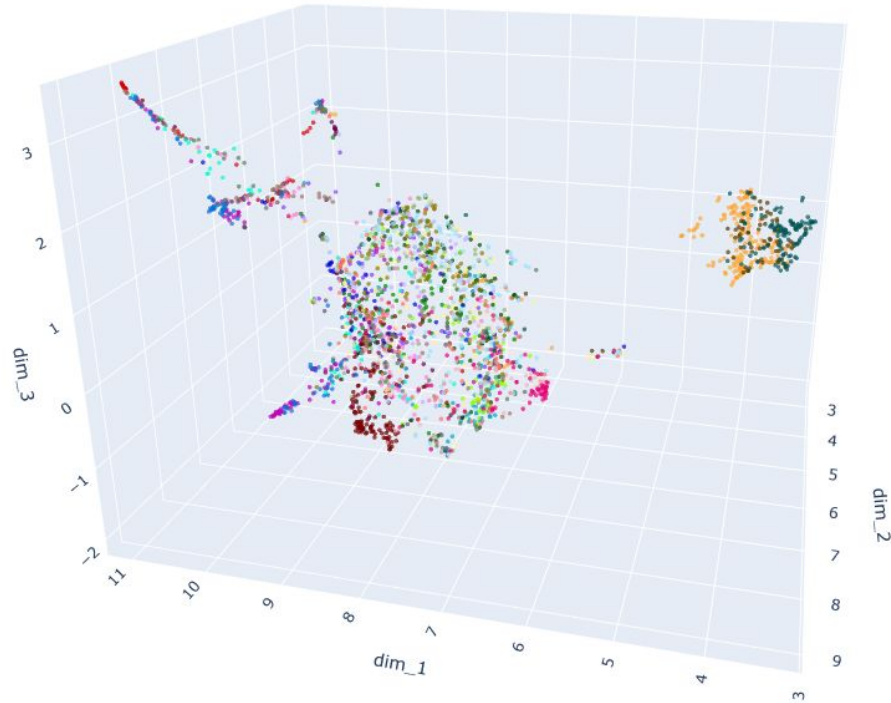
ResNet-50, t-SNE 2D



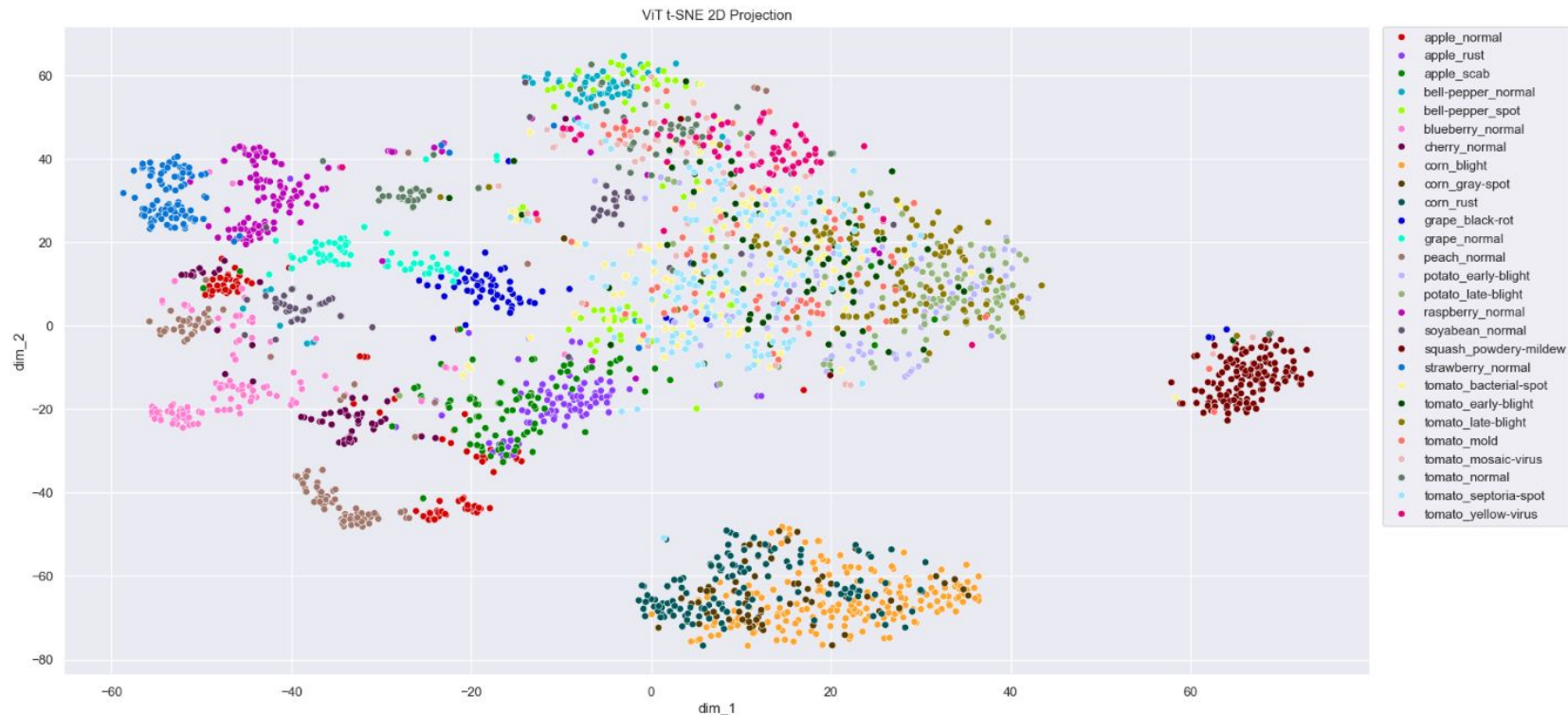
ResNet-50, UMAP 2D



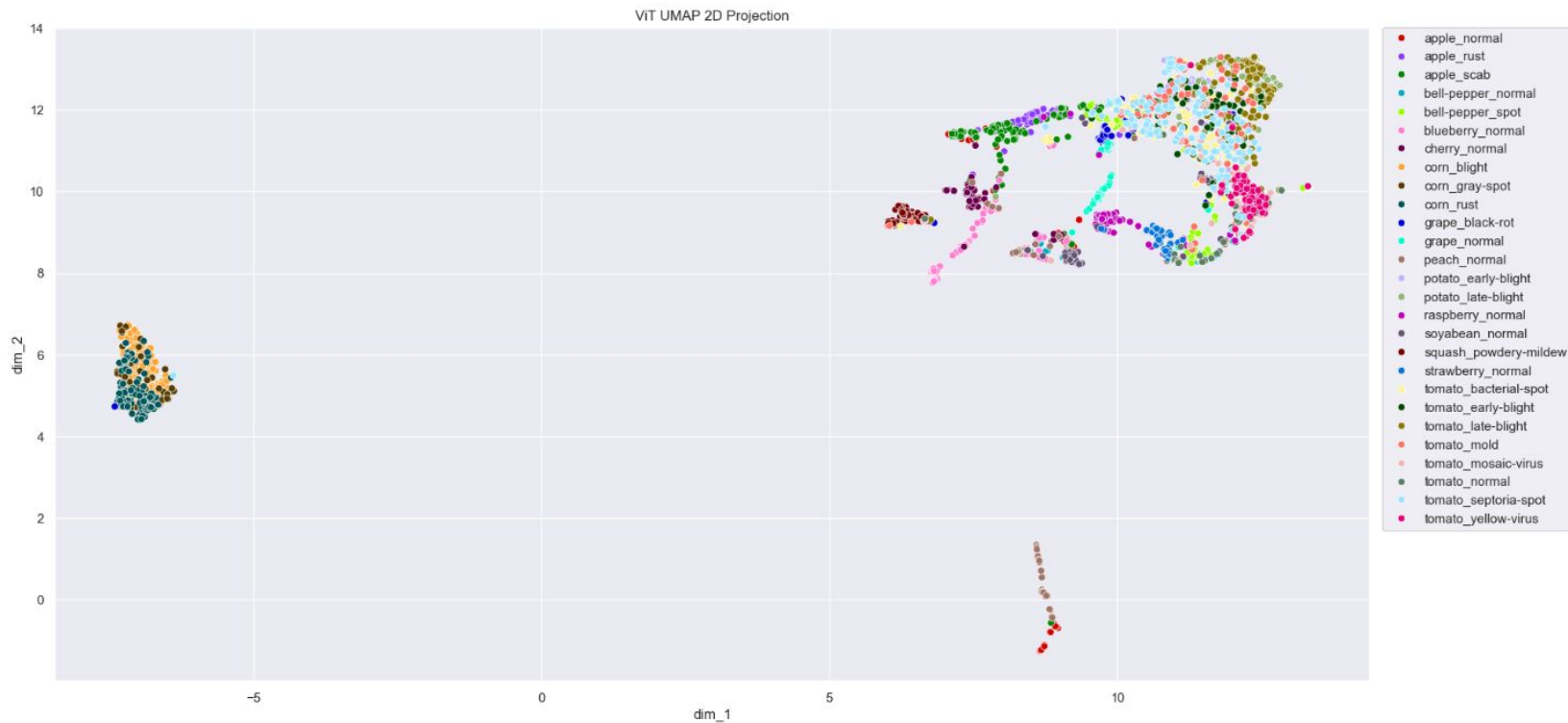
ResNet-50, UMAP 3D



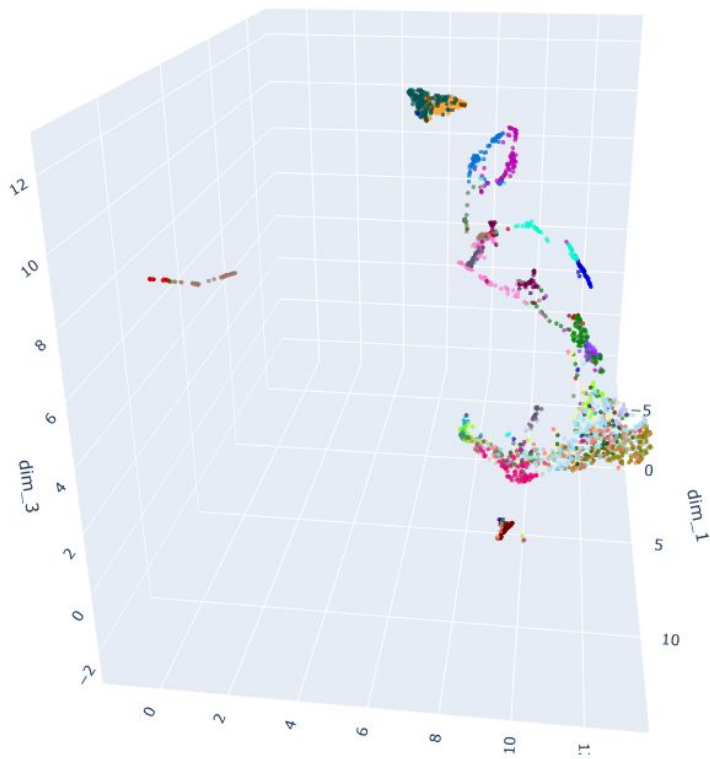
Google ViT, t-SNE 2D



Google ViT, UMAP 2D



Google ViT, UMAP 3D



Clustering

1

DBSCAN

2

HDBSCAN

3

Mahalanobis K-Means

Clustering

Clustering algorithms are applied to visualize what data points the models are positioning **semantically close together** and to understand whether the models are able to tell classes apart.

In order to evaluate the performance of the clustering algorithms employed, difference metrics have been used:

- **Internal Validation**
 - *Density Based Cluster Validation Score, DBCV*, for **density-based** clustering
 - *Silhouette Score*, for **K-means like** clustering
- **External Validation**
 - *Adjusted Rand Index*

Like with manifold learning algorithms, clustering algorithms aswell require some hyperparameters to be tuned. Optuna has been employed on internal validation scores and UMAP results.

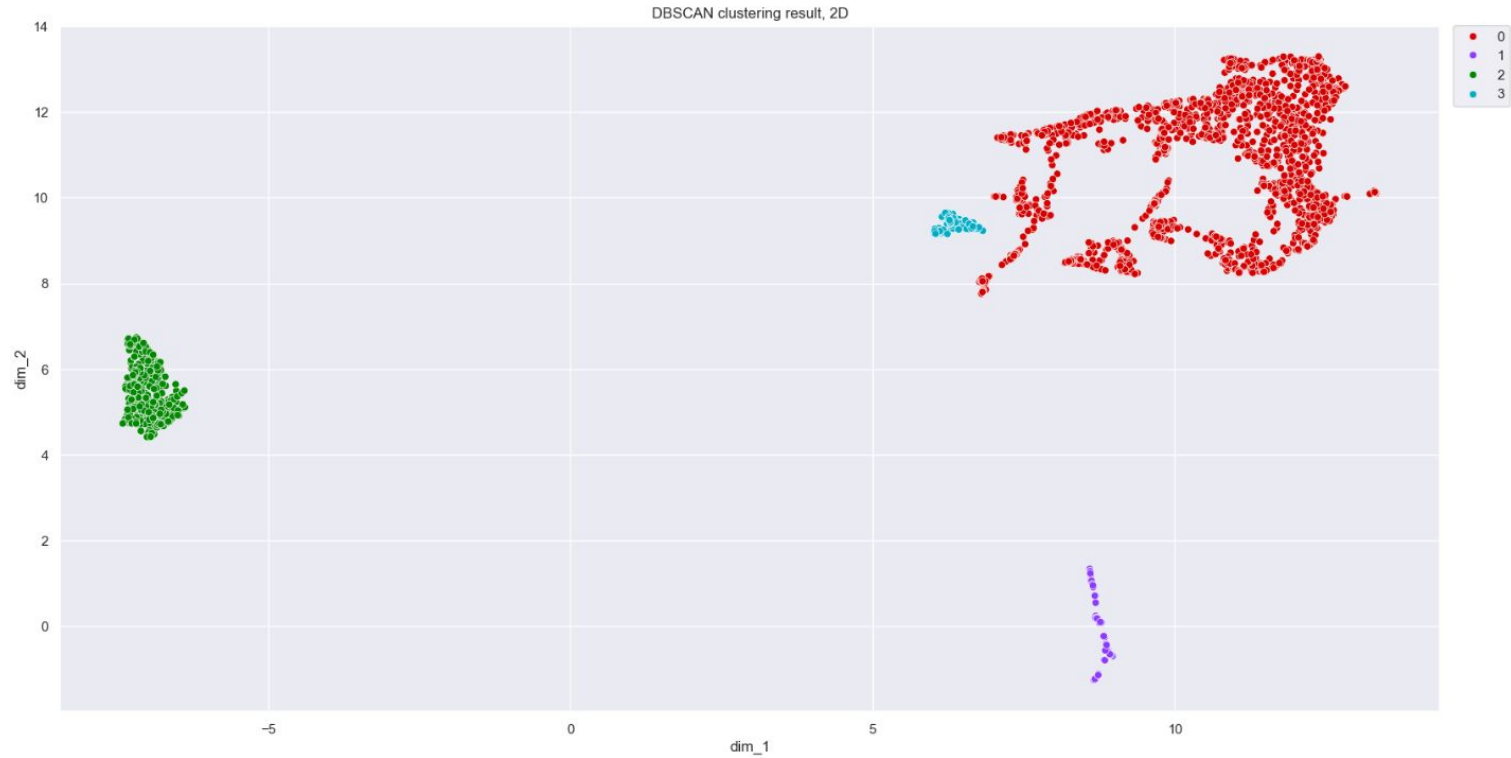
DBSCAN

The idea behind DBSCAN is that of defining a **density factor** to merge data points into clusters.

The **density** of a data point is defined by the number of points **t** that are within a certain radius **e** from the point, itself included. This ends up defining three possible points: **core**, **border** and **noise**.

The way DBSCAN is implemented implies that **all clusters** must be characterized by the **same density**.

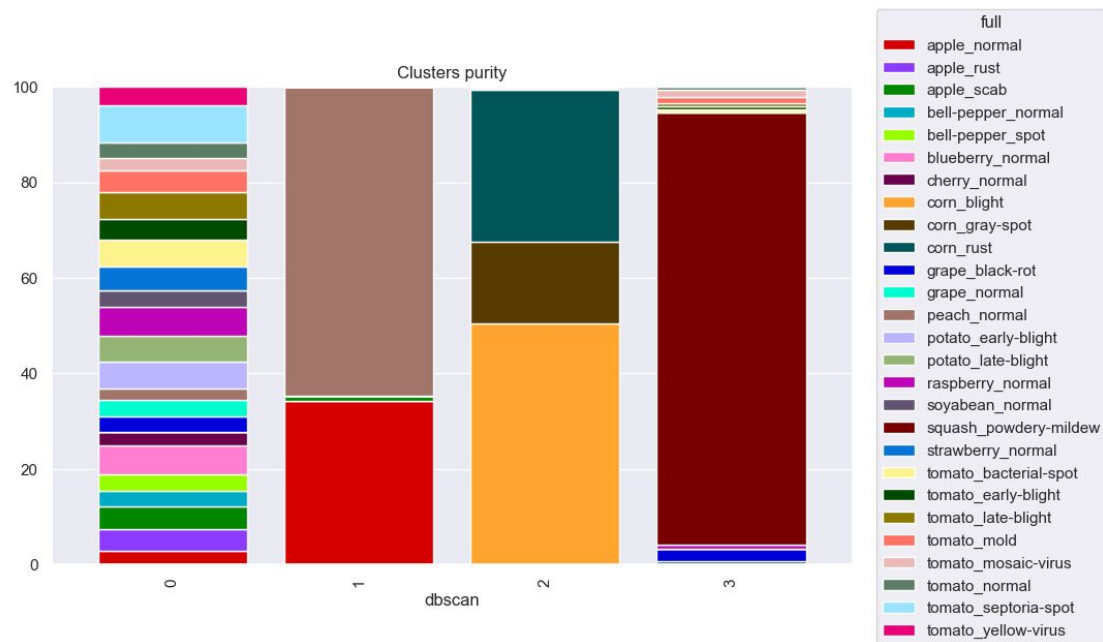
DBSCAN



DBSCAN Clusters Purity

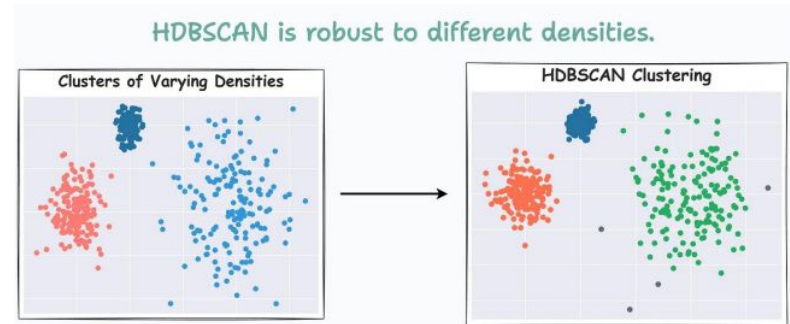
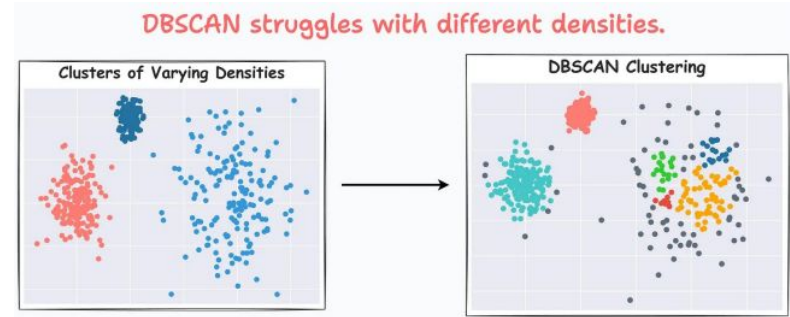
The *Internal Validation Score* (DBCV) is quite high, ~**0.83**.

The *External Validation Score* (ARI) is really low, **0.0492**

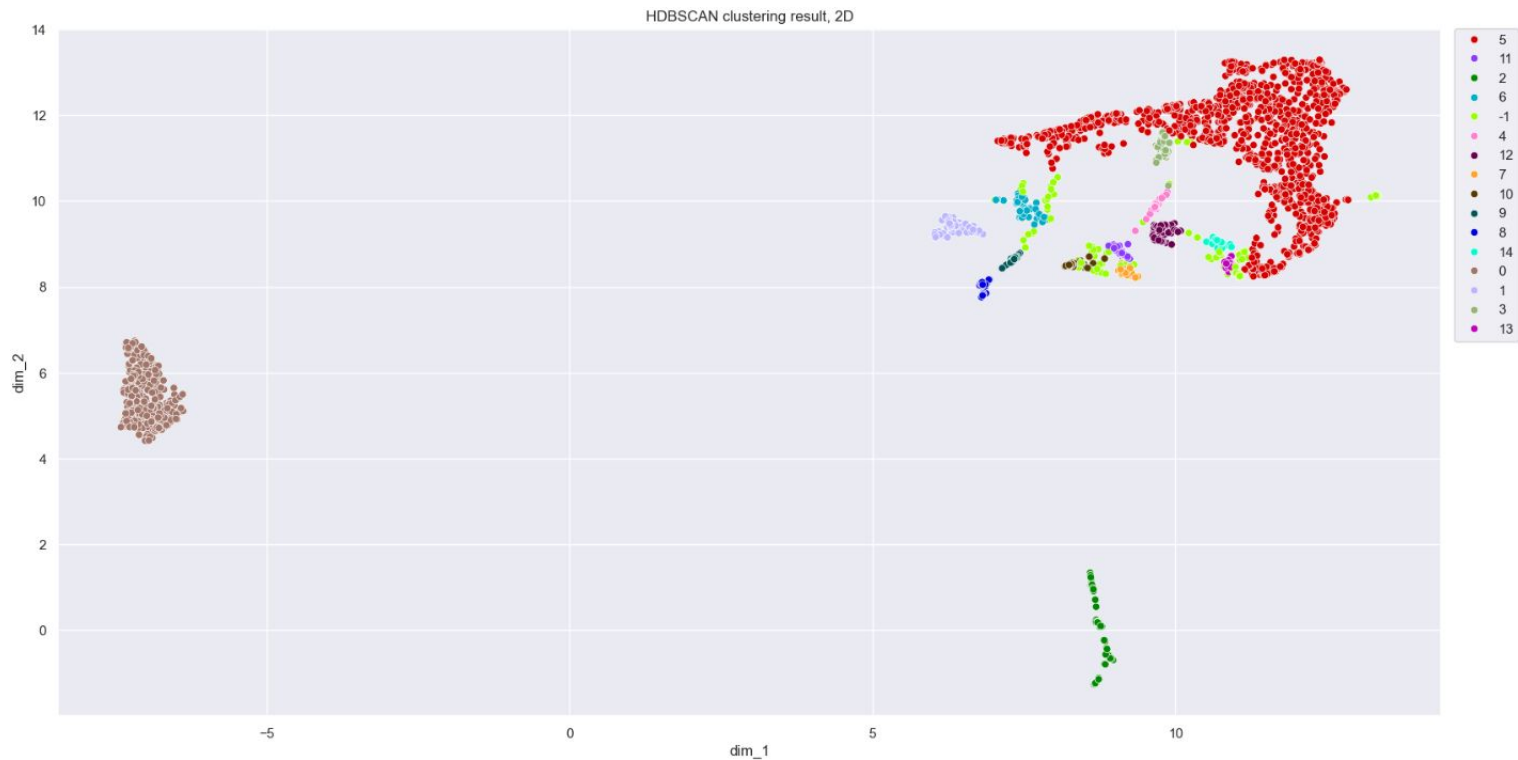


HDBSCAN

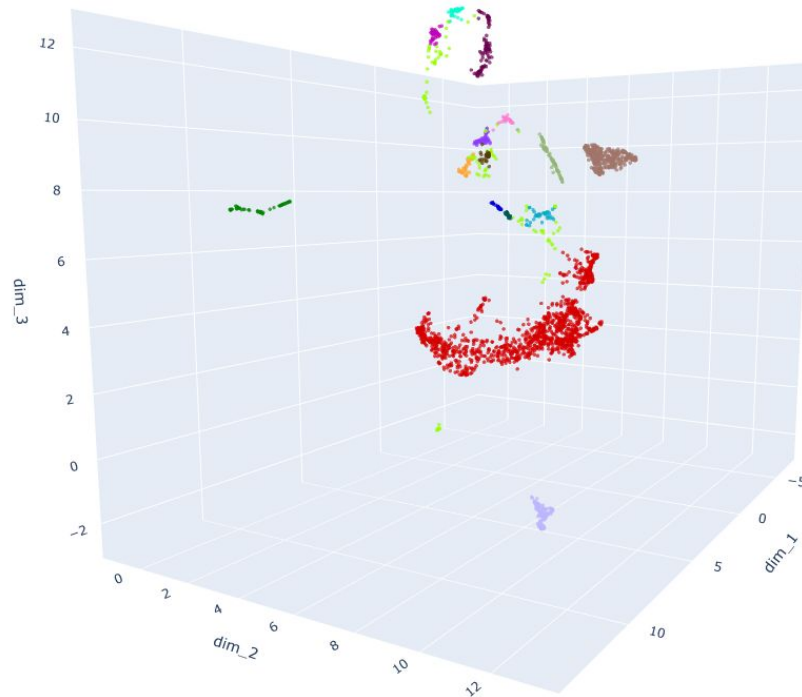
HDBSCAN algorithm improves on DBSCAN by having a **hierarchical** approach to the creation of clusters, allowing for different possible values of **densities** instead of a single fixed one



HDBSCAN, 2D



HDBSCAN, 3D

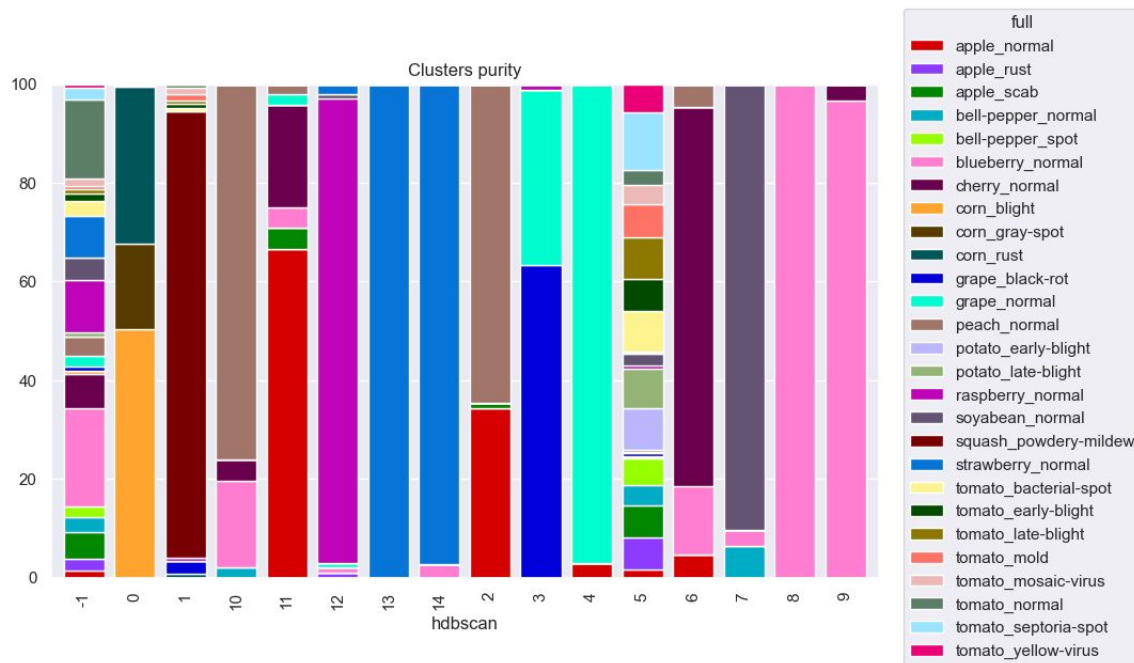


HDBSCAN Clusters Purity

The *Internal Validation Score* (DBCV) is not as high as with DBSCAN, **~0.57**.

The *External Validation Score* (ARI) is still quite low, **0.1417**.

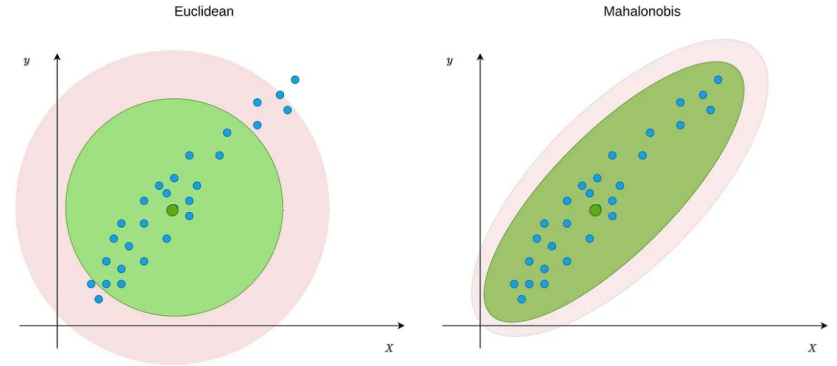
HDBSCAN manages way better the **small** clusters.



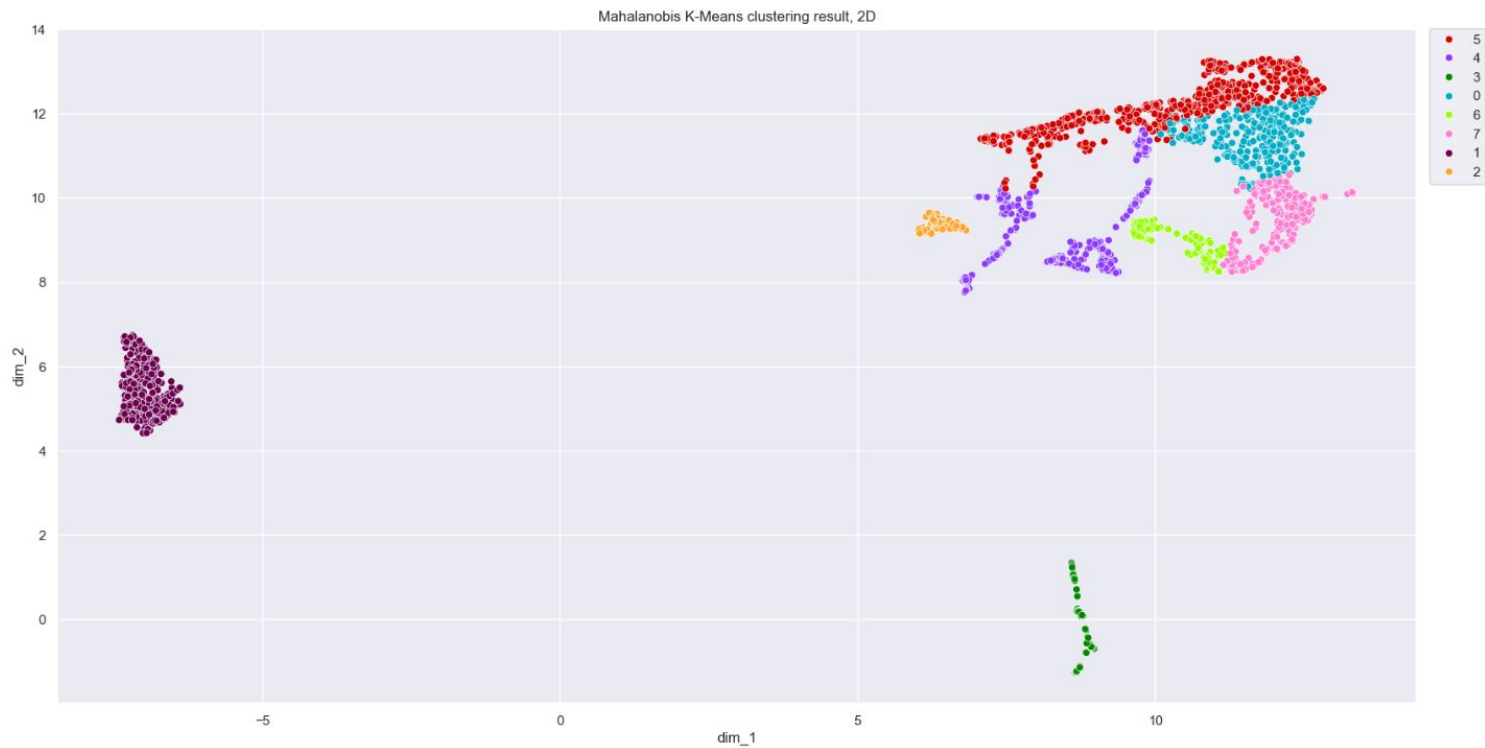
Mahalanobis K-Means

Mahalanobis K-Means is an implementation of the K-Means algorithm where the distance metric used is the **Mahalanobis distance** which takes into account the **covariance matrix**.

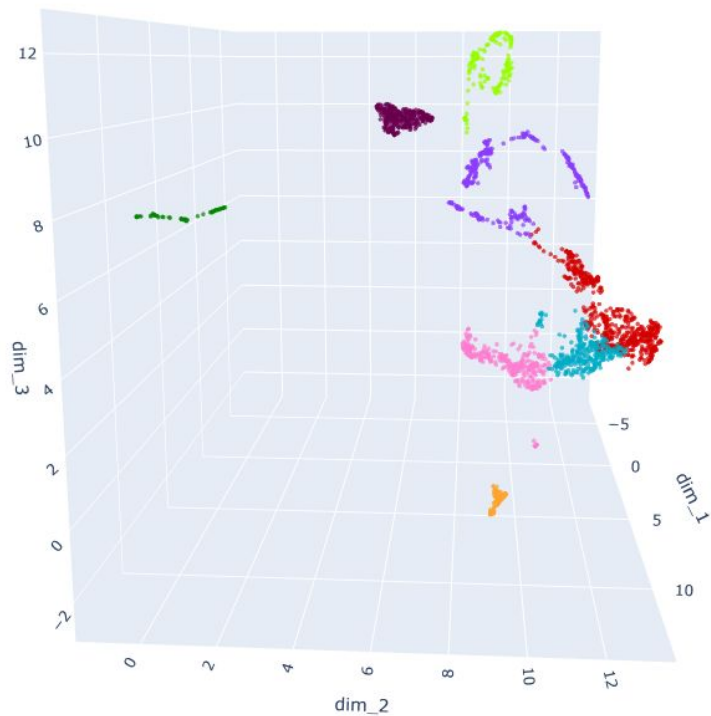
This allows for the algorithm to create **ovally** shaped clusters instead of the **spherical** ones that the **Euclidean distance** creates.



Mahalanobis K-Means, 2D



Mahalanobis K-Means, 3D

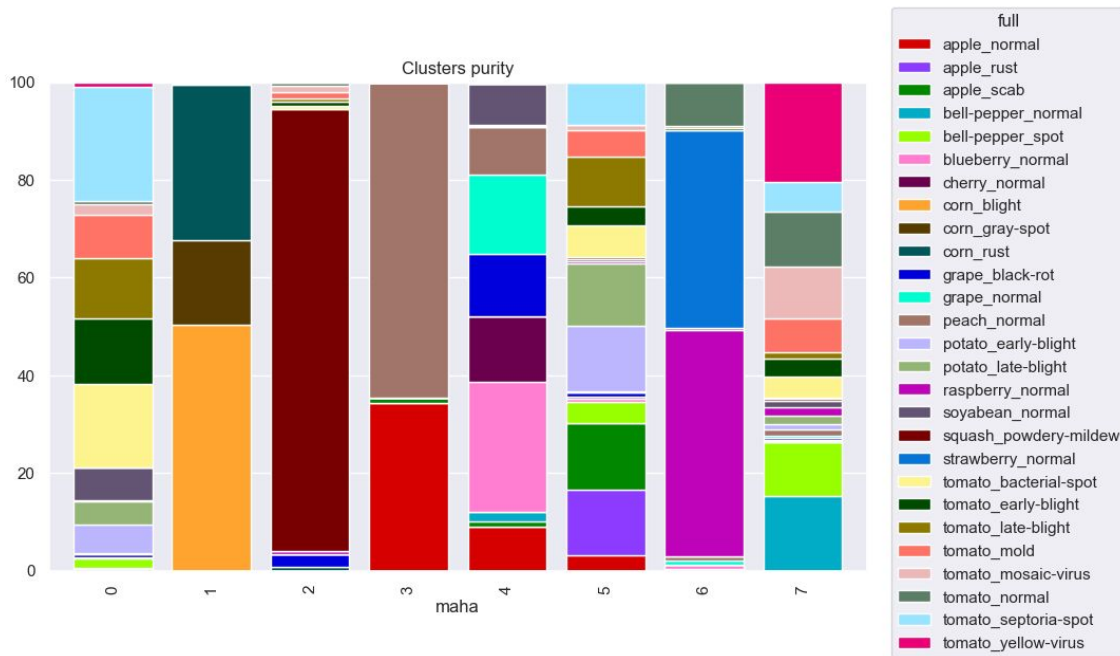


Mahalanobis K-means Clusters Purity

The *Internal Validation Score* (Silhouette) is mid-to-high, ~**0.64**.

The *External Validation Score* (ARI) is still low, **0.2484**.

Mahalanobis K-Means appear to be able to better dissect the big blob into smaller portions.



Models Fine Tuning & XAI

1

Fine Tuning

2

XAI

Models Fine Tuning

The results obtained so far by ResNet-50 and Google ViT are coming from the knowledge that these models have extracted from their training process, respectively on **ImageNet-1k** and **ImageNet-21k**.

These two dataset do not contain specifically the classes that the problem at hand is characterized by, Corn being the only exception since it is present in **ImageNet** (*"Ear of Corn"*).

In order to improve the performances of the two models, it is necessary to perform a **fine tuning** process that allows the models to be **trained again and on top of their already existing knowledge** with the images from PlantDoc. This will allow the models to extract the details found specifically in this dataset and to gain more knowledge on this specific problem.

Data Augmentation

In order to provide **as many different images** to the models such that they will be able to better generalize on unseen data, **data augmentation** is performed. This process has to be carefully done in order **not to “confuse”** the models with images that do not make sense:

- | | | |
|-------------------------|---|--|
| Color | → | Images cannot have their color altered since ending up with color combinations that characterize another disease could immensely confused the models. |
| Random crops | → | Images cannot be randomly cropped since they could end up losing the area of the plant that provides information on the disease |
| Compress/Stretch | → | Images cannot be compressed or stretched in one dimension since they could end up losing the proportions information provided on the disease |
| Mirroring | → | Images can be mirrored since plants images do not carry information on horizontal or vertical symmetries |
| Rotation | → | Images can be rotated since plants images do not carry information on specific directions |

Data Augmentation, Color

Both images are leaves of **Apple** with the first being affected by **Rust** and the second by **Scab**.

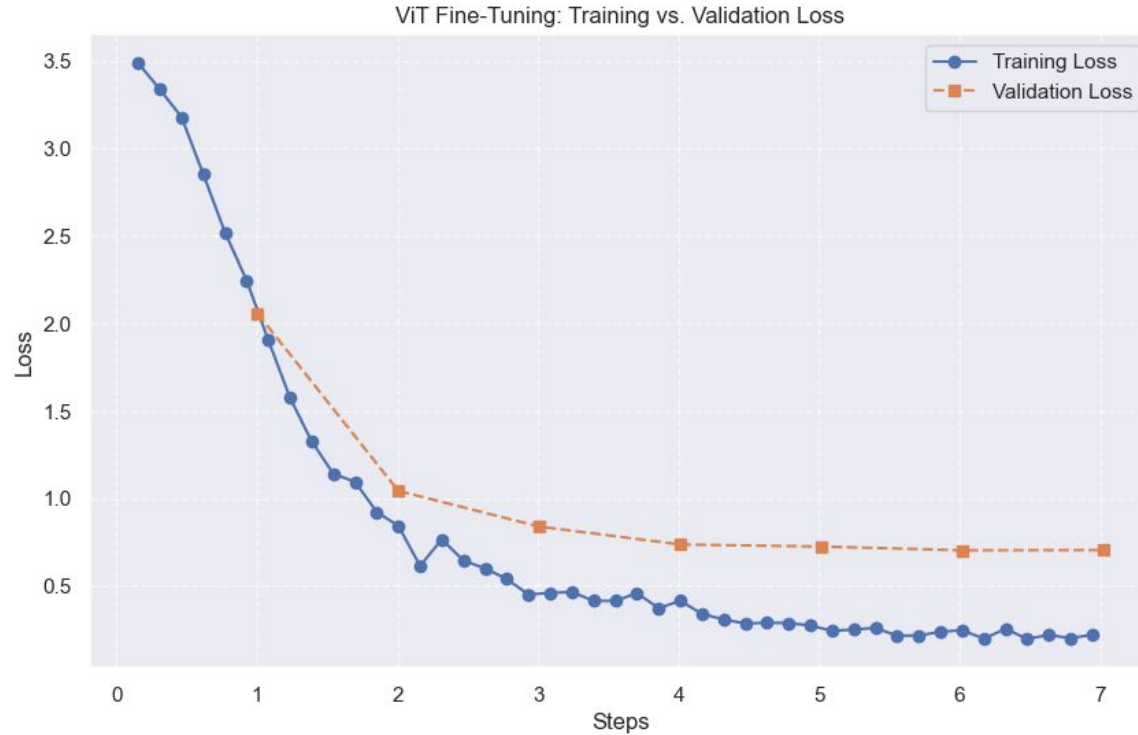
It is possible to notice how in case **Color** augmentation was applied and the **orange-ish** spots turned **brown-ish** it would have resulted in creating confusion for the models.



Model Training, ResNet-50



Model Training, Google ViT



XAI

Due to the complex nature of the two models, a **XAI tool** has been employed to provide more transparency on **what** parts of images have been used by the models to learn.

The tool that has been used is **Saliency Maps**.

Saliency Maps, ResNet-50

Original Plant Leaf



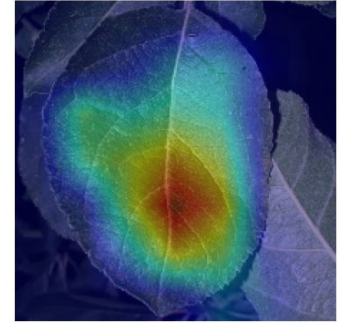
Grad-CAM (Class Index: 0)



Original Plant Leaf



Grad-CAM (Class Index: 2)



Original Plant Leaf



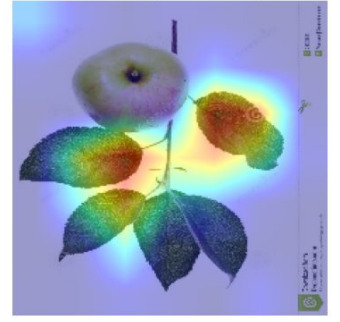
Grad-CAM (Class Index: 0)



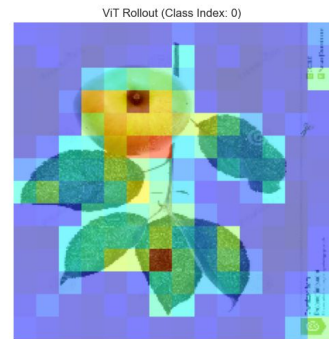
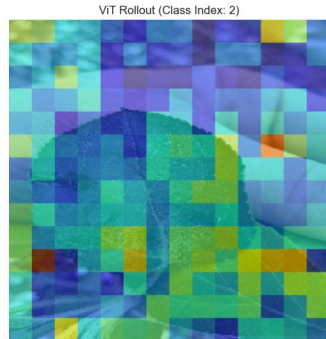
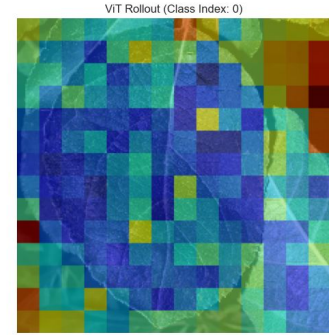
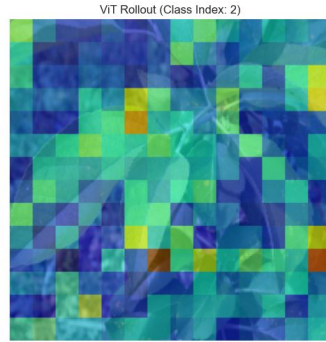
Original Plant Leaf



Grad-CAM (Class Index: 0)



Saliency Maps, Google ViT



Classification

1

ResNet-50

2

Google ViT

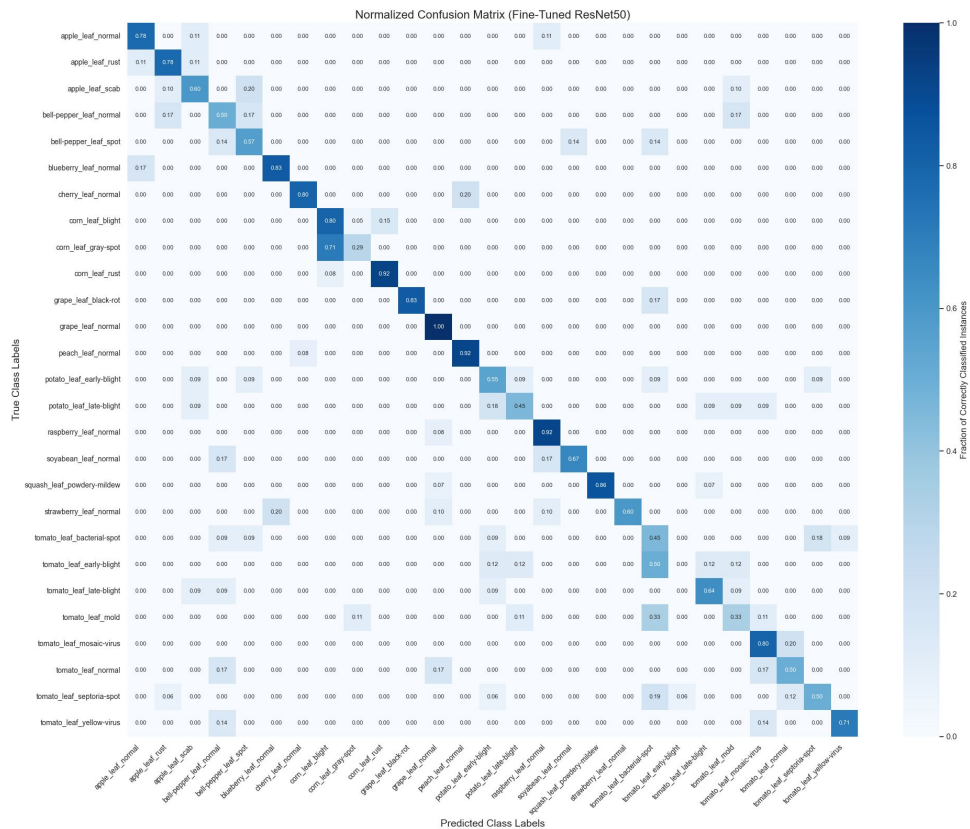
Classification

Due to the classes imbalance present in the dataset, the selected metric to compare the performances of the two models is the **F1-score** such that the minority classes still play a meaningful role in the result.

Classification, ResNet-50

	precision	recall	f1-score	support					
					potato_leaf_late-blight	0.6250	0.4545	0.5263	11
					raspberry_leaf_normal	0.7857	0.9167	0.8462	12
					soyabean_leaf_normal	0.8000	0.6667	0.7273	6
					squash_leaf_powdery-mildew	1.0000	0.8571	0.9231	14
					strawberry_leaf_normal	1.0000	0.6000	0.7500	10
					tomato_leaf_bacterial-spot	0.2778	0.4545	0.3448	11
					tomato_leaf_early-blight	0.0000	0.0000	0.0000	8
					tomato_leaf_late-blight	0.7000	0.6364	0.6667	11
					tomato_leaf_mold	0.3750	0.3333	0.3529	9
					tomato_leaf_mosaic-virus	0.5000	0.8000	0.6154	5
					tomato_leaf_normal	0.5000	0.5000	0.5000	6
					tomato_leaf_septoria-spot	0.7273	0.5000	0.5926	16
					tomato_leaf_yellow-virus	0.8333	0.7143	0.7692	7
					accuracy			0.6654	260
					macro avg	0.6504	0.6518	0.6408	260
					weighted avg	0.6722	0.6654	0.6593	260
apple_leaf_normal	0.7000	0.7778	0.7368	9					
apple_leaf_rust	0.7000	0.7778	0.7368	9					
apple_leaf_scab	0.5455	0.6000	0.5714	10					
bell-pepper_leaf_normal	0.3333	0.5000	0.4000	6					
bell-pepper_leaf_spot	0.4444	0.5714	0.5000	7					
blueberry_leaf_normal	0.8333	0.8333	0.8333	12					
cherry_leaf_normal	0.8000	0.8000	0.8000	5					
corn_leaf_blight	0.7273	0.8000	0.7619	20					
corn_leaf_gray-spot	0.5000	0.2857	0.3636	7					
corn_leaf_rust	0.8000	0.9231	0.8571	13					
grape_leaf_black-rot	1.0000	0.8333	0.9091	6					
grape_leaf_normal	0.6364	1.0000	0.7778	7					
peach_leaf_normal	0.9167	0.9167	0.9167	12					
potato_leaf_early-blight	0.5000	0.5455	0.5217	11					

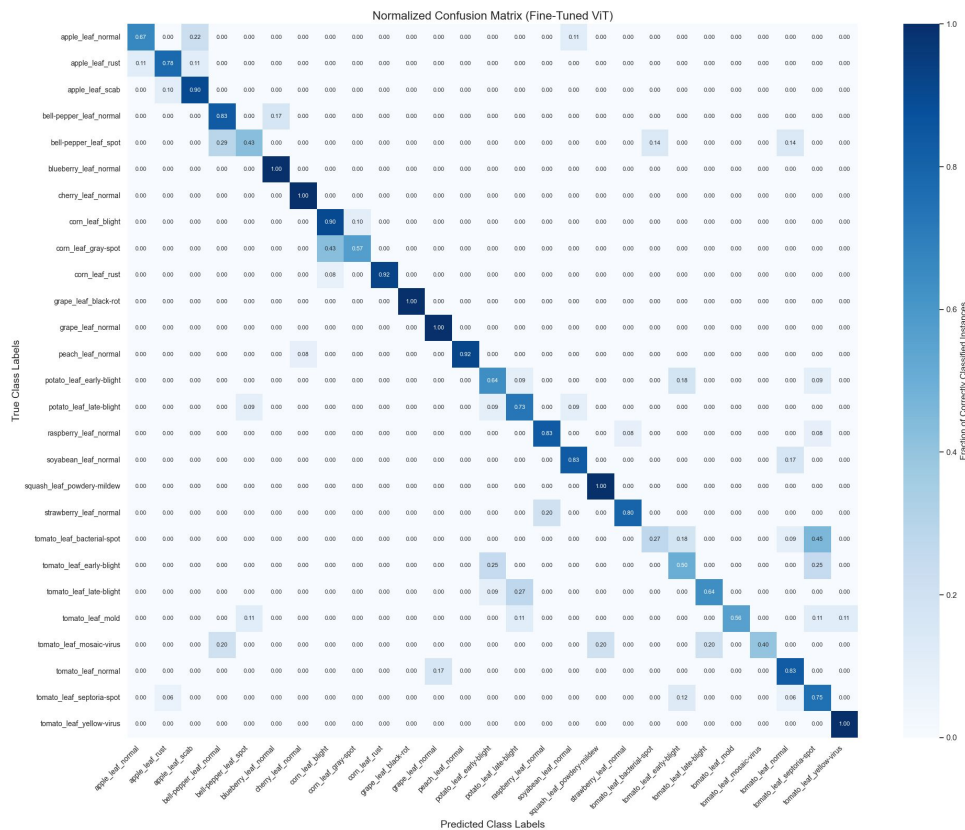
Classification, ResNet-50



Classification, Google ViT

	precision	recall	f1-score	support					
apple_leaf_normal	0.86	0.67	0.75	9	potato_leaf_late-blight	0.62	0.73	0.67	11
apple_leaf_rust	0.78	0.78	0.78	9	raspberry_leaf_normal	0.83	0.83	0.83	12
apple_leaf_scab	0.75	0.90	0.82	10	soyabean_leaf_normal	0.71	0.83	0.77	6
bell-pepper_leaf_normal	0.62	0.83	0.71	6	squash_leaf_powdery-mildew	0.93	1.00	0.97	14
bell-pepper_leaf_spot	0.60	0.43	0.50	7	strawberry_leaf_normal	0.89	0.80	0.84	10
blueberry_leaf_normal	0.92	1.00	0.96	12	tomato_leaf_bacterial-spot	0.75	0.27	0.40	11
cherry_leaf_normal	0.83	1.00	0.91	5	tomato_leaf_early-blight	0.40	0.50	0.44	8
corn_leaf_blight	0.82	0.90	0.86	20	tomato_leaf_late-blight	0.88	0.64	0.74	11
corn_leaf_gray-spot	0.67	0.57	0.62	7	tomato_leaf_mold	1.00	0.56	0.71	9
corn_leaf_rust	1.00	0.92	0.96	13	tomato_leaf_mosaic-virus	1.00	0.40	0.57	5
grape_leaf_black-rot	1.00	1.00	1.00	6	tomato_leaf_normal	0.56	0.83	0.67	6
grape_leaf_normal	0.88	1.00	0.93	7	tomato_leaf_septoria-spot	0.55	0.75	0.63	16
peach_leaf_normal	1.00	0.92	0.96	12	tomato_leaf_yellow-virus	0.88	1.00	0.93	7
potato_leaf_early-blight	0.64	0.64	0.64	11					
					accuracy			0.78	260
					macro avg	0.79	0.77	0.76	260
					weighted avg	0.80	0.78	0.77	260

Classification, Google ViT



Comparison

1

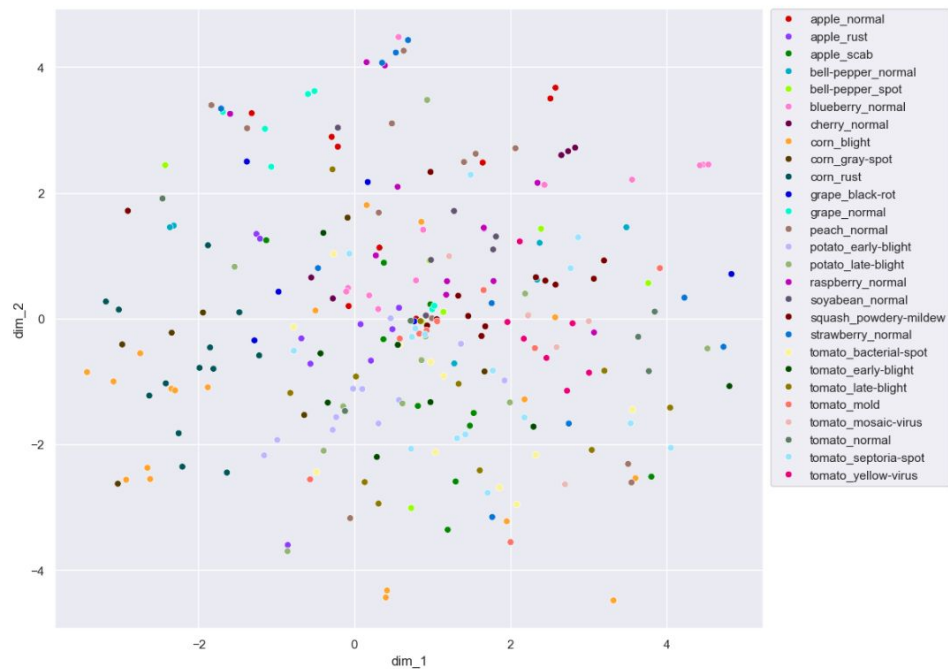
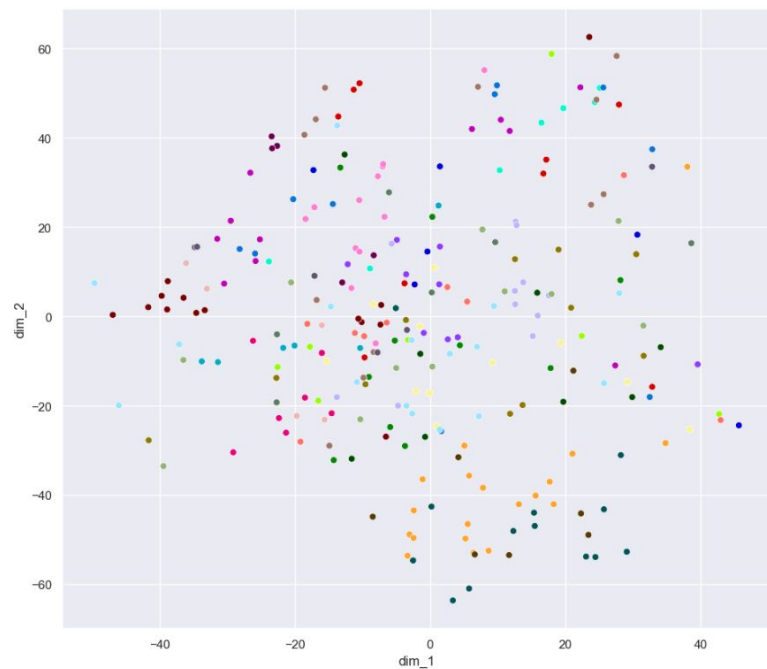
ResNet-50

2

Google ViT

ResNet-50, t-SNE 2D

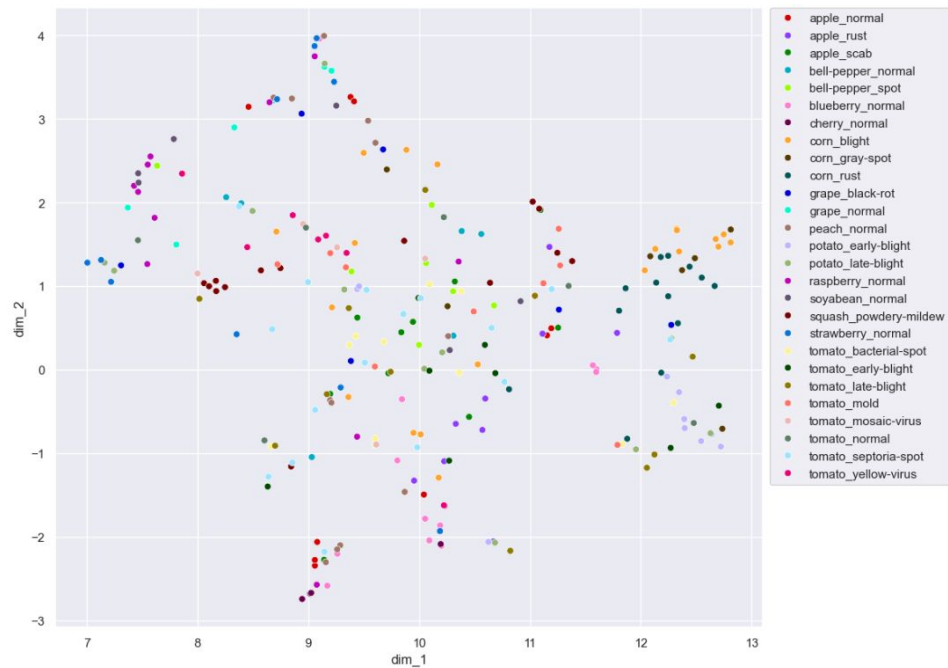
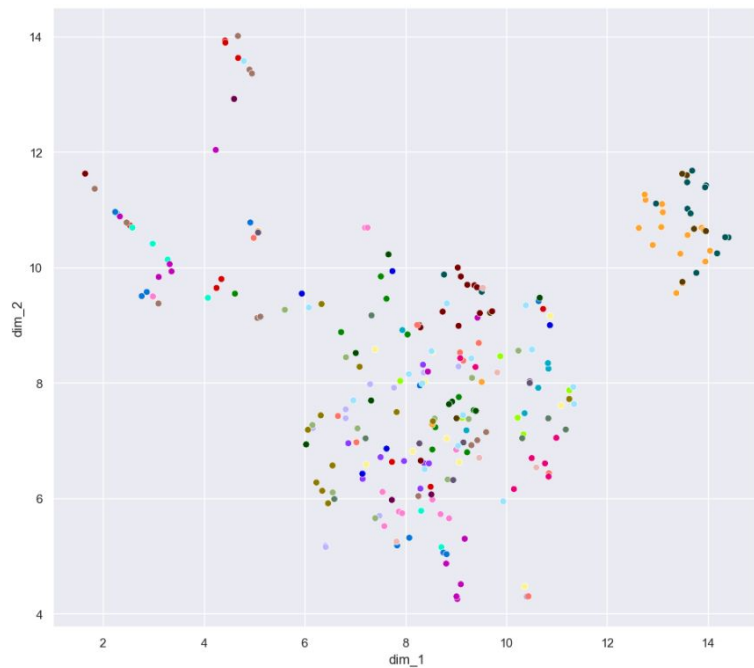
Pre fine tuned vs. fine tuned ResNet-50 t-SNE 2D



Comparison

ResNet-50, UMAP 2D

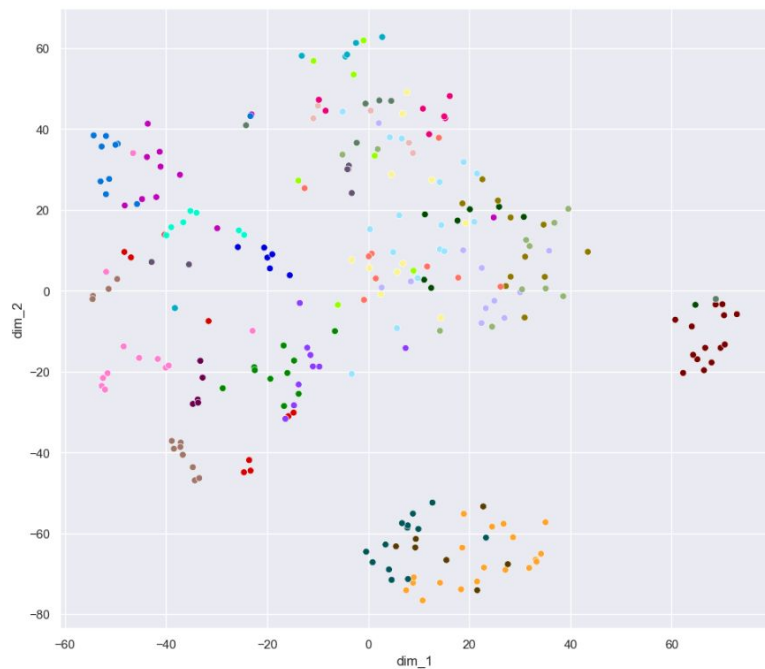
Pre fine tuned vs. fine tuned ResNet-50 UMAP 2D



Comparison

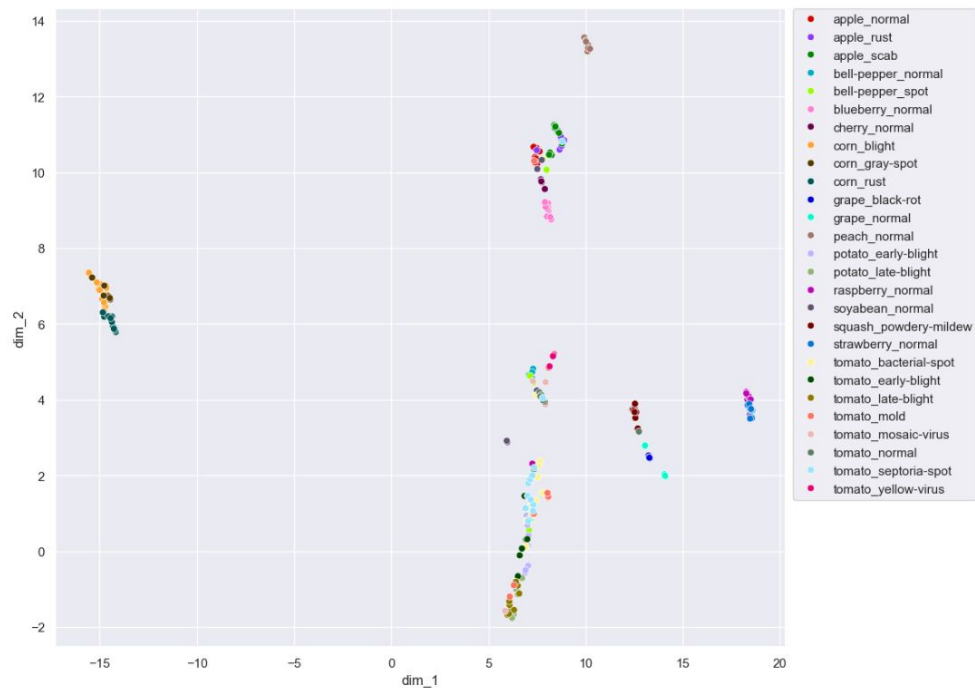
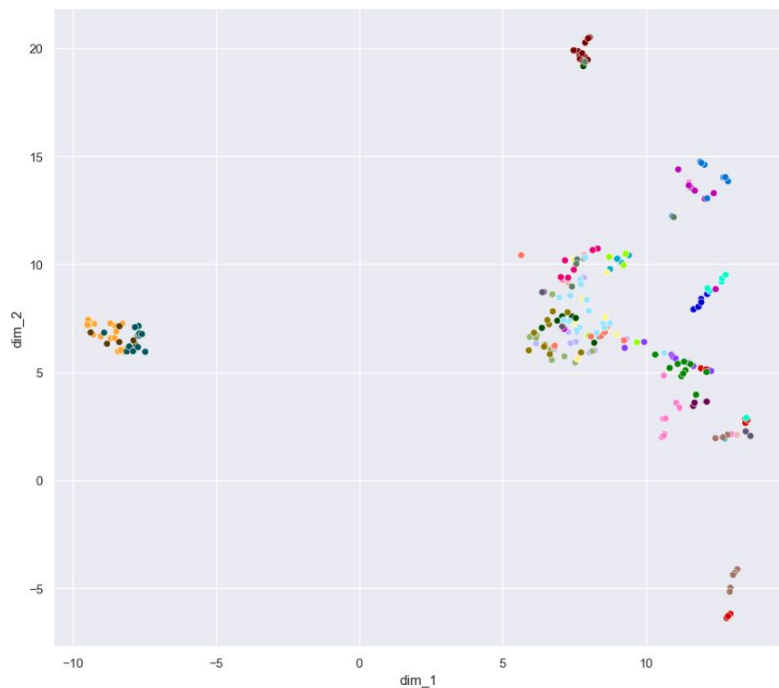
Google ViT, t-SNE 2D

Pre fine tuned vs. fine tuned ViT t-SNE 2D

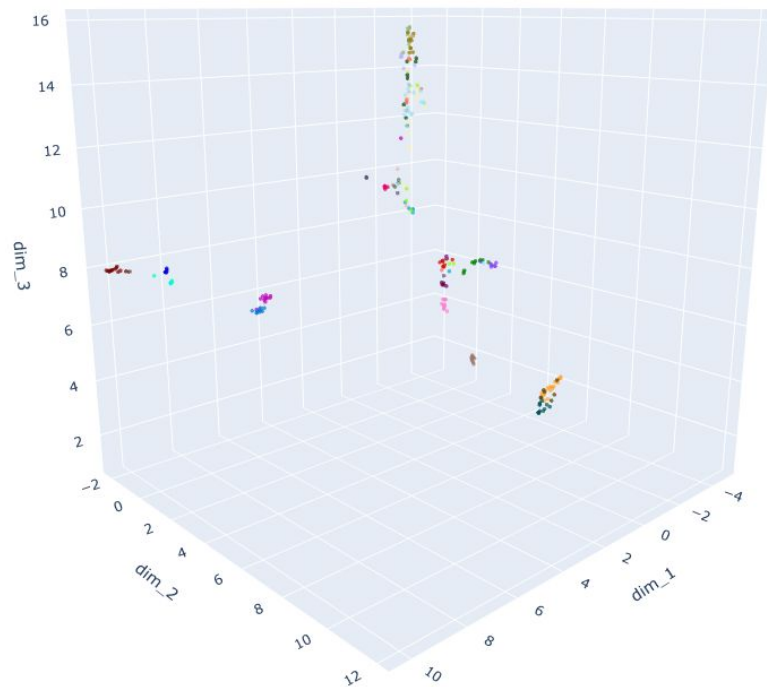
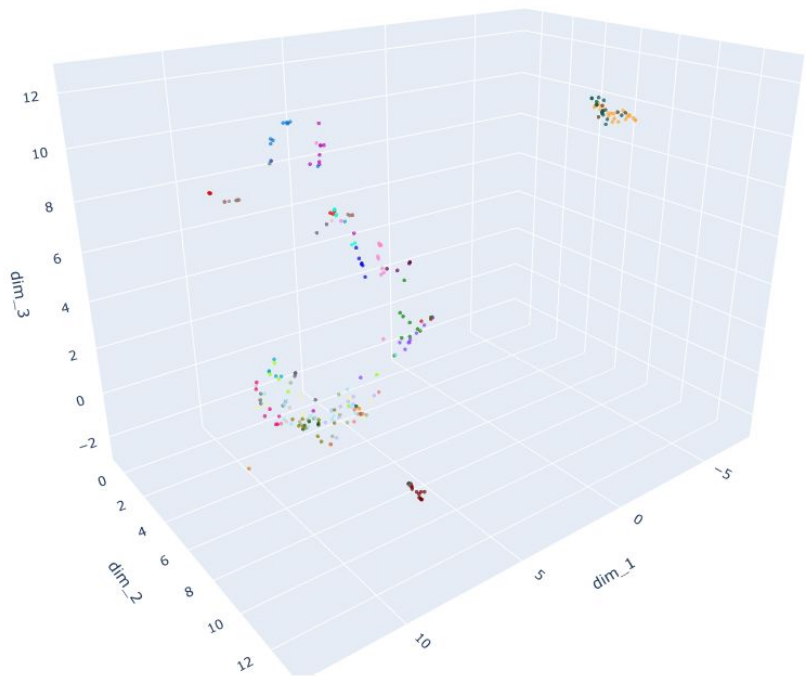


Comparison

Google ViT, UMAP 2D



Google ViT, UMAP 3D



Conclusions

The **data exploration** step showed the heterogeneity of the data: some being **real-world** taken and others being **lab-controlled**. This is not an issue but rather a way to enrich the knowledge of the models that are going to be fine tuned, making them **more robust** to different types of images.

The **embedding extraction** step, thanks to PCA and both t-SNE and UMAP, highlighted how ResNet-50 appears to provide a more confused representation of the data compared to the Google ViT Base Patch16-224.

Furthermore, employing **clustering** helped to underline what classes were semantically positioned close to each other from both models and what could be the most troublesome classes to separate.

After **fine tuning** both models it was possible to perform a classification task that showed, again, how the ViT proved to be a better model for classification purposes for this problem.

Finally, after running again the embedding extraction process, both models showed an **improved representation** of data.

Current challenges:

- Due to the nature of the problem, certain plant species are rather hard to separate due to being part of the same family such as the case of both Tomatoes and Potatoes coming from the **Solanaceae** family.